What an Auxiliary Cue Must Carry: LiDAR Body Pose for Pedestrian Trajectory Forecasting

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***Abstract*—Gait and facing appear to encode pedestrian intent that a position history lacks, which makes body pose an attractive forecasting input. On the Waymo Open Motion Dataset (WOMD), with SMPL poses recovered from LiDAR, we find that whether pose pays depends on the track it is added to and on what the pose pipeline was itself given. The decisive comparison is correct pose against the same architecture fed another pedestrian's pose. On WOMD's tracks, offboard auto-labels refined with future frames, it returns $+0.04\%$ (p=0.93) at our largest scale, bounded to $\pm 1.7\%$. With the history a causal onboard tracker would report, own pose beats a stranger's by 5.3% (p=0.002) at 30 cm of detection noise. Three quarters of that gain is the label returning through the body: the pipeline's motion-prior fit matches the gait to the offboard track. With poses that never saw the track the gain is 1.3% (p=0.10), and 3.5% (p=0.010, uncorrected) on pedestrians with at least 200 LiDAR returns. An oracle through the same branch converts the agent's own future bearing, so the pathway is not the obstacle. Before integrating a modality, measure what it carries beyond the model's existing inputs at the quality a vehicle will have, and check which labels its own pipeline consumed.**

***Index Terms*—pedestrian trajectory prediction, human pose estimation, LiDAR, autonomous driving, offboard labels**

I. Introduction

Predicting where a pedestrian will go is a safety-critical function of an automated vehicle, and body pose is an obvious candidate input. A person's stance, gait phase and facing direction plausibly reveal intent before their position changes [1], [2], and modern perception stacks already produce human mesh or keypoint estimates. Several recent systems condition trajectory forecasts on pose [3]–[6].

This paper reports that, on the labelled tracks of the Waymo Open Motion Dataset (WOMD), pose conditioning does not measurably improve pedestrian trajectory accuracy, and that the reason is the track it is added to. WOMD's boxes are offboard auto-labels, refined with the whole track, future frames included [7], [8]. Against that track the body restates what the model already knows. Against the history a causal onboard tracker would report, the same body appears to pay, and most of that gain is the offboard label returning through the pose pipeline, which fits the gait to the track. Poses that never saw the track appear to pay only where the sensor resolves the body (Fig. 1).

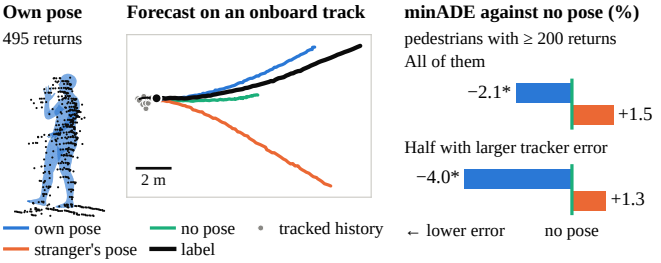


Fig. 1. Under a simulated onboard tracker (30 cm noise), a pedestrian's own SMPL pose from LiDAR lowers minADE against no pose by 2.1% on the well-resolved pedestrians of Table III, and by 4.0% on the half with the larger tracker velocity error. A stranger's pose does not. The poses never saw the track, asterisks mark $p<0.05$, uncorrected, and the example is from Fig. 4, selected on the effect, each model's median seed.

A controlled null on offboard tracks. We measure pose conditioning against a shuffled-pose control: the same architecture receiving a different pedestrian's body motion. This isolates information content from the capacity cost of an extra branch. The control is null at three training scales, under two integration styles and across three pose encodings (6D joint rotations, agent-frame root, and eight frame-invariant descriptors). An oracle arm, which replaces the body with a signal encoding the agent's own future, improves minADE by 34.8% through the same branch, so the null is a property of what pose adds to these tracks, not of the plumbing.

An onboard-quality track, and a label leak. We replace WOMD's observed history with what a causal tracker would report, per-frame detection noise filtered by a forward-only Kalman filter, and repeat the content test. Own pose now beats a stranger's by 5.3% at 30 cm of detection noise. Three quarters of that is not body information. The pose pipeline ends in a motion-prior fit whose loss matches the gait to the box track, so the fitted body states the clean speed the tracker lost. Rerun on the pre-fit poses, the gain is 1.3% and no longer significant. On well-resolved bodies it is 3.5% and grows with LiDAR returns, a result that rests on one stratification.

Within them the gain follows the tracker's error and not the pedestrian's manoeuvre.

What the pathway converts, and what it does not. We corrupt that oracle with a controlled angular error to ask what kind of information the branch turns into accuracy. At 25k scenes every error level we trained converts, including one uniformly worse than an input the model already has, so the ladder does not set an accuracy a cue must reach. It separates a precision term with a 3.0° half-life from a slowly decaying term that is future-side mode selection. Body facing supplies neither: it is a noisier copy of the box heading where pedestrians move, and reaches 41.2° where they do not, the case velocity cannot cover.

II. Related Work

Trajectory forecasting on WOMD is dominated by anchor- and query-based transformers. We build on MTR [9], which is representative and strong. Human mesh recovery from LiDAR is maturing [10], with motion priors used to complete sparse observations [11]–[13]. Intent-focused work on pedestrian datasets [14], [15] targets crossing decisions rather than metric trajectories. Kress et al. [6] report pose-based gains for vulnerable road users from a research vehicle, at close range and with camera-derived pose. Methodologically our closest precedent is Makansi et al. [16], who find by feature attribution that trajectory predictors draw almost nothing from an input the field had assumed was central.

Relation to prior positive results. Pose conditioning has reported real gains, and our mechanism predicts where they come from rather than contradicting them. Social-Transmotion [3] lowers ADE on the synthetic JTA dataset from 0.99 to 0.89 with ground-truth 3D pose and from 0.40 to 0.36 on JRDB with estimated pose. Social-Pose [4] reports about 25% on JTA ($1.20 \rightarrow 0.90$) and on JRDB. Salzmann et al. [5] report $\approx 10\%$ from onboard-estimated keypoints at ranges up to 7 m, for agents with little position history. A short history is a poor track and 7 m is a well-resolved body, the one condition under which our track-free poses appear to pay (Sec. IV-E). Le et al. [17] localise the JRDB gain to lower-body 3D keypoints (13% ADE).

The closest published setting to ours already reports approximately our result. Social-Pose also evaluates on Pedestrians and Cyclists in Road Traffic, the one vehicle-captured dataset there, and with ground-truth 3D pose ADE moves from 0.58 to 0.57 m, by 0.01 m or less for all five architectures tested, against the $\approx 25\%$ the same encoder delivers on JTA. Two properties separate the remaining positive settings from an AV stack. The first is range and sensor. JRDB is an indoor and campus robot dataset in which pedestrians are metres away and pose is camera-derived, whereas WOMD pedestrians are observed from 2 to 75 m and often return a few dozen LiDAR points. The second is the baseline. Those predictors consume positions, not a box heading, so body facing is new information there and partly duplicated here. None reports a shuffled-pose arm.

III. Setup

A. Data and a data-integrity failure

We use WOMD [7] with SMPL [18] poses recovered from LiDAR by the Waymo-3DSkelMo pipeline [19], which fits per-frame bodies with LiDAR-HMR [10] and completes them using AMASS-trained motion priors [11]–[13]. Poses cover the 11 observed frames, and the prediction horizon is 8 s at 10 Hz. In every arm the model sees the map and the pedestrians that carry poses. Vehicles, cyclists and the ego vehicle are not in its input. Every predicted agent carries at least one valid pose frame, and 86.7% of observed frames carry one.

One property of that pipeline bears on Sec. V: when a pedestrian is moving and the raw LiDAR estimate of root rotation deviates from the past-trajectory direction by more than 90°, the pipeline snaps the root to the direction of travel. Body facing is therefore partly trajectory-derived for moving pedestrians. This reinforces the redundancy we report rather than weakening it, and it does not reach the stationary-onset analysis, whose events are all below 0.3 m/s. A second property bears on Sec. IV-E: the motion-prior stage fits each sequence against the box track, so the completed body is not independent of that track. We call the per-frame LiDAR-HMR output that precedes it the pre-fit pose (Fig. 3).

Our first pass at this study was invalid. The export keyed pose clips by a scene-local object identifier, so clips collided across scenes and nearly every training sequence carried another pedestrian's body. A per-scene re-export (299,220 scenes, 1.81M sequences) cut the median error between root heading and travel direction from 41.5° to 11.4° and raised the gait–speed correlation from $r{=}0.26$ to 0.58. The failure is silent: training converges and losses look normal. Every number here is from the corrected export.

B. Arms and the content test

Three arms share one architecture (Fig. 2). The no-pose arm receives trajectories and map. The pose arm adds the pedestrian's own SMPL sequence, 144 joint-rotation values per frame with no translation, through a cross-attention encoder with sinusoidal positional encoding. The shuffled arm receives another pedestrian's sequence, assigned once over the whole split by a fixed derangement, with identical parameter count, optimiser and input dimensionality. The root orientation enters in world coordinates while the trajectories are agent-centric, so this arm does not state facing relative to the model's frame. The agent-frame-root encoding does, and its content test is null too ($+0.68\%$, $p{=}0.49$, $n{=}3$, MDE 4.3 pp). A fourth arm, the oracle, replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, leaving the branch and its masks untouched. It shows the pathway can carry information at all (Sec. IV-C). Every arm with a pose branch, the stranger and the oracle included, is also trained to predict the future pose sequence it was given, under an auxiliary loss (a geodesic rotation term and a mixture likelihood term, weight 0.1 each). The no-pose arm has no such loss.

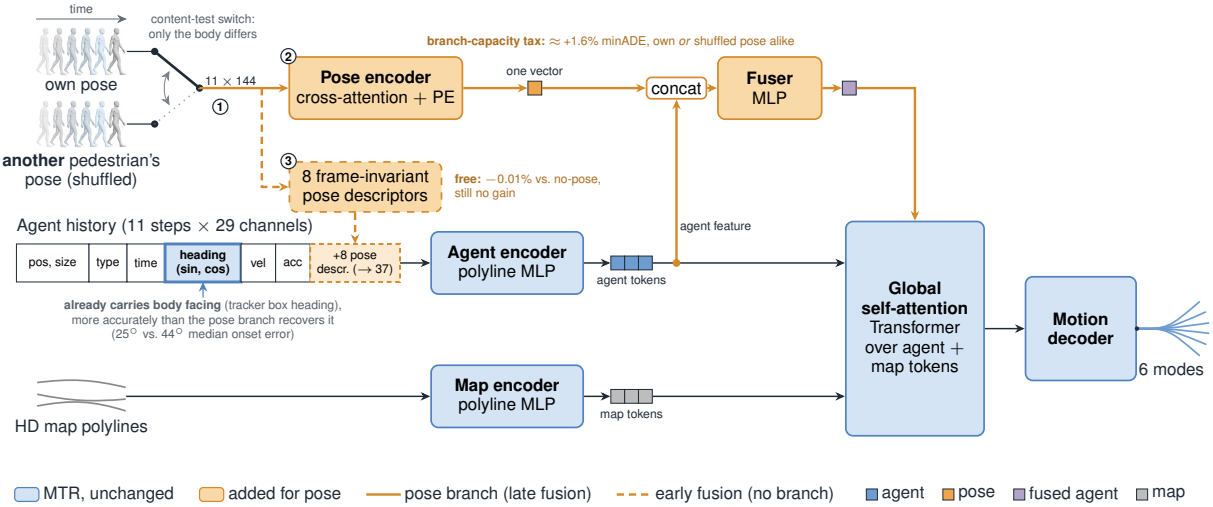


Fig. 2. The two arms differ only in whose body enters the model. The content-test switch (1) feeds the predictor either the pedestrian's own SMPL sequence or another pedestrian's, holding parameters, optimiser and input dimensionality fixed. Blue: MTR, unchanged. Orange: added for pose. The agent-history strip shows that the labelled box heading is already an input. Solid orange (2): the pose branch (late fusion). Dashed (3): early fusion (Sec. IV-B).

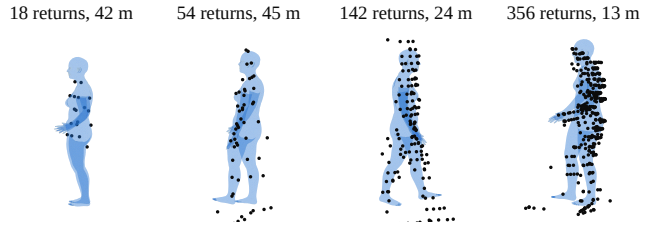


Fig. 3. LiDAR returns (black) and the pre-fit SMPL body estimated from them (blue), at one metric scale. One scored pedestrian per stratum of mean returns over the observed second, cut at 30, 100 and 200: the pedestrian at the stratum median, in its frame closest to that mean. Titles give the frame's returns and range.

Pose versus no-pose confounds information with branch capacity and the auxiliary loss, while pose versus shuffled holds both fixed and isolates content. We call the latter the content test and treat it as decisive. Relative differences are $(A - B)/B$, so negative favours A.

Inference is over seeds, the unit the treatment is applied to. Unless flagged, every comparison between trained arms is a paired t-test on per-seed differences.

C. Protocol

Every arm trains for 30 epochs with AdamW (two scenes per GPU with at most 24 predicted pedestrians each, weight decay 0.01, gradient-norm clipping 1.0) under a step schedule that halves the learning rate at epochs 10, 20 and 25, and the decoder predicts $K{=}6$ modes. The learning rate is chosen per scale from a grid run separately for each arm: $1{\times}10^{-4}$ at 579 scenes, $5{\times}10^{-5}$ at 5k and $2.5{\times}10^{-5}$ at 25k, where every arm peaks at the same interior optimum. Validation is fixed at 5,171 scenes. Of their 17,218 predicted pedestrians, the 16,486 with at least one valid future step are scored. Headline numbers are taken at each run's best validation epoch over a fixed evaluation schedule, and Sec. IV-B reports what changes under a fixed-epoch rule. Official WOMD mAP needs the full retained prediction set, which our per-epoch evaluation does not store, so we report minADE, minFDE and miss rate at 2 m. minADE averages over a pedestrian's valid future steps and minFDE is taken at the last valid one, and 36% of the scored pedestrians have a full 8 s future. Code, configs, per-seed metrics and the analysis scripts are released with the paper.

TABLE I

CONTENT TEST ACROSS TRAINING SCALE, VALIDATION FIXED. THE LEARNING RATE IS SELECTED PER SCALE BY A GRID RUN SEPARATELY FOR EVERY ARM, AND ALL ARMS PEAK AT THE SAME VALUE (SEC. III-C). PERCENTAGES ARE DIFFERENCES OF ARM MEANS, AND p-VALUES ARE PAIRED t-TESTS OVER PER-SEED DIFFERENCES. NEGATIVE FAVOURS POSE.

Training scenes	579	5k	25k
Pedestrians	6,348	31,302	142,807
Seeds	5	3	5
minADE (m)			
No-pose	0.4599	0.4176	0.3716
Pose	0.4620	0.4138	0.3778
Shuffled	0.4589	0.4181	0.3777
Pose vs. no-pose	+0.45%	−0.90%	+1.66%*
Content test	+0.66%	−1.02%	+0.04%
p (content)	0.52	0.56	0.93

*$t(4){=}10.3$, $p{=}5{\times}10^{-4}$. The shuffled arm pays the same cost, +1.63%, $t(4){=}2.8$, $p{=}0.048$.

IV. RESULTS

A. The null on offboard tracks, and its bound

Table I trains all three arms at three scales. Data alone improves the baseline by 19.2%, while the content test is null at every scale, in minADE, in minFDE (+0.45%, −0.28% and +0.33%, $p \geq 0.61$) and in miss rate at 2 m, where it is +2.2% at 25k ($p{=}0.21$) and +4.3% fine-tuned ($p{=}0.19$): null, and with its sign against pose on all three metrics. At 80% power

the minimum detectable content effect is 3.5% at 579 scenes (n=5), 7.6% at 5k (n=3) and 2.2% at 25k (n=5). The 5k scale is weak evidence on its own, and the weight falls on 25k, the largest scale and the most sensitive. There the paired seed-level interval on the content difference is $[-1.59\%, +1.70\%]$, so the data exclude a content effect beyond roughly 1.7% in either direction, and the largest benefit the interval allows is no larger than the 1.6% cost of the branch that would carry it.

After pretraining a trajectory-only backbone on full WOMD and fine-tuning at 579 scenes (n=5), the content test is $+0.88\%$ (p=0.18), again with its sign against pose.

B. Two effects that are real, and are not pose

A branch-capacity tax. At 25k both pose arms are worse than no-pose ($+1.66\%$ at $p=5\times10^{-4}$ and $+1.63\%$ at p=0.048) and indistinguishable from each other. An extra input branch costs $\sim1.6\%$ whether it carries the right body or the wrong one. Both arms also train with the auxiliary pose loss, which we did not separate from the branch at this scale.

This is an operating-point statement: it is measured at each run's best validation epoch. The sign of a pose-versus-no-pose comparison is not stable under a fixed late-epoch rule: at epoch 30 pose is 8.0% better with n=3, where the no-pose arm had degraded further, and 0.9% worse with n=8, both n.s. and neither resolved at these seed counts. What is stable is that the comparison moves with the stopping rule, so we never compare architecturally different arms at a fixed late epoch. The content test is not significant under either rule, though it is noisier at epoch 30 (-5.9%, -4.7% and -1.7% at the three scales, p=0.16, 0.07 and 0.60). Replacing the 144-d branch with eight frame-invariant descriptors fused into the existing agent channels removes the cost (-0.01% versus no-pose at p=0.99, and -1.67% versus the branch at p=0.12, n=3) without creating a benefit: pose can be carried for free, and still does not pay. Fed an uncorrupted oracle at 25k scenes this pathway converts it on every seed, by -31.2% (p=0.0002) against the branch's -34.5%, so a channel it costs nothing to carry is still a channel it reads.

Tuning dominates the treatment. Across the 25k grid the learning rate moves minADE by up to 16.7% against a content signal of $\sim0.05\%$, so every arm is tuned separately.

C. The branch is not inert

A null measured through one pathway invites the objection that the pathway is dead, and the shuffled control does not dispose of it: a branch that transmits nothing would also fail to tell its own body from a stranger's.

We therefore ran the positive control. The oracle arm of Sec. III is predictive by construction, and it keeps the branch, its parameter count, its input dimensionality and its per-frame validity masks exactly as the pose arm has them, so the oracle is visible only where a real pose would have been.

Over three seeds at 579 scenes the oracle arm changes minADE by -34.8% against no pose ($t(2) = -81.4$, $p=1.5\times10^{-4}$) and by -35.6% against the shuffled arm. The oracle reads the future and is not a usable model. It calibrates the pathway: the branch converts information into accuracy at roughly twenty times the magnitude of the bound our content test places on pose. The null is therefore a statement about what SMPL pose contains, not about whether the architecture can use what it is given.

D. Where a benefit could still hide

On an eight-seed replication of the content test at 30 epochs, scored on a held-out half of validation (about 8,040 agents), it is null at every horizon and least favourable at the shortest ($+0.48\%$ at 2 s, -0.04% at 8 s). It is null in all 20 tests across quartiles and top deciles of constant-velocity residual, heading variation, speed range and peak acceleration, on the 6,764 of those agents for which the four are defined. It is also null for standing-then-moving pedestrians, the subpopulation built to favour it (n=1,809 pedestrians, $+0.38\%$, epoch-30 checkpoints on an expanded validation split, p=0.80 by a pedestrian-resampling bootstrap).

It is also null where the body is best measured, which is where a pose-quality explanation predicts an effect should appear. Stratifying the 25k content test by LiDAR return count gives -0.37% ($[-1.87, +1.14]$) on the 2,372 agents with at least 200 returns and -0.08% on the 10,902 with at least 30, and stratifying by range gives -0.14% ($[-1.61, +1.33]$) inside 10 m. No stratum shows an effect, at n=5 throughout.

Across the more than forty content tests this study reports outside the onboard-track ladder of Sec. IV-E, two reach $p<0.05$, and both favour the stranger. In the stationary subpopulation of the heading ablation, correct pose is $+1.8\%$ worse than a stranger's (n=8, p=0.011), and in the stationary subset of the frame-rotation control at $\sigma=30^\circ$ it is $+1.3\%$ worse (n=3, p=0.049, Sec. V-B). At that multiplicity one or two such results are expected by chance, and their direction is against a pose mechanism rather than for one. The tests of Sec. IV-E that favour pose are a different kind of result: they grow with the noise and agree in sign on every seed, and most of what they measure is a label, not a body.

E. An onboard-quality track, and a label leak

Everything above is measured on WOMD's tracks, and those are offboard auto-labels. An offline network refines every box with the whole track, future frames included [7], [8], and the velocity the baseline receives is exactly the backward difference of those refined positions at all 954,751 pedestrian time steps we checked. The observed history is as clean as the labeller could make it. A deployed stack has a causal tracker instead, so we simulate one. Each observed position receives detection noise $\mathcal{N}(0, s^2)$ per axis, drawn once rather than per epoch, and a forward-only constant-velocity Kalman filter, started at rest (acceleration noise 1 m/s^2, initial velocity uncertainty 1.5 m/s), turns the detections into the positions and velocities the model sees. The box heading and the future targets are untouched. At prediction time, measured on 675 pedestrians, the tracker is off by 7.5 cm and 0.23 m/s in median at s=10 cm, and by 21 cm and 0.46 m/s at 30 cm, against a median pedestrian speed of 1.1 m/s.

TABLE II

DEGRADED INPUTS AT 25K SCENES, n=3 SEEDS PER ARM FOR THE FRAME ROTATION AND n=5 FOR TRACK NOISE. LOSS IS THE NO-POSE BASELINE AGAINST ITS CLEAN CELL ON THE SEEDS BOTH SHARE, AND NEGATIVE FAVOURS POSE. THE FRAME ROTATION LEAVES THE CLEAN HEADING READABLE, SO IT IS NOT AN ONBOARD-QUALITY HEADING (SEC. V-B). THE FITTED POSES SAW THE OFFBOARD TRACK, SO THEIR TRACK-NOISE ROWS OVERSTATE WHAT THE BODY ADDS. THE LAST ROWS REPEAT THE 30 CM LEVEL ON PRE-FIT POSES, OVERALL AND BY MEAN LIDAR RETURNS.

Degraded input	Loss	Content	p	vs. no-pose
None	—	+0.41%	0.60	+1.97%
Agent frame rotated by a fixed error $\mathcal{N}(0, \sigma^2)$				
σ=15°	+5.8%	−0.14%	0.90	−0.39%
σ=30°	+12.6%	−0.11%	0.93	+1.34%
Track, causal tracker on detection noise s, fitted poses				
s=10 cm	+14.1%	−2.08%	0.005	−1.21%
s=30 cm	+32.3%	−5.32%	0.002	−4.06%
Track at s=30 cm, pre-fit poses				
All pedestrians	+32.3%	−1.34%	0.10	+0.36%
$<$ 30 returns	—	−0.34%	0.72	+1.82%
$\geq$ 200 returns	—	−3.47%	0.010	−1.99%

With the pipeline's fitted poses the null breaks (Table II). At 30 cm own pose beats a stranger's by 5.32% (p=0.002, 95% CI $[-7.3, -3.4]$) and beats no pose by 4.06% (p=0.004), over five seeds. All five agree in sign (−2.7 to −6.5%), and the mean holds at every fixed late epoch (−5.2 to −5.8% at epochs 26, 28 and 30). At 10 cm it is −2.08% against shuffled (p=0.005) and −1.21% against no pose (p=0.07). It sits where gait would put it. On the 8,757 pedestrians walking at 0.8 to 1.6 m/s it is −3.3% (p=0.008) at 10 cm and −6.8% (p=0.0002) at 30 cm, while on the 3,336 standing at prediction time it is significant at neither. It read to us as a stride restoring the speed the tracker lost.

Most of it is the label. The fitted poses are not independent of the track. The motion-prior stage [12] is a latent optimisation whose loss predicts the root translation from the local body motion and matches it to the box track, so the fit pulls the gait toward a stride that reproduces the offboard displacement. The model input holds joint rotations only and covers only the 11 observed frames, so auditing the input does not reveal it. A check without training does. On the 11,588 validation pedestrians with a pose on every observed frame, gradient-boosted trees predict the clean one-second speed from body pose alone, root excluded, with R^2=0.94 from the fitted pose and 0.44 from the pre-fit one. On moving pedestrians with fewer than 30 LiDAR returns it is 0.78 against less than zero: the fitted body knows a speed the sensor could not have shown it. Added to a regressor on the 30 cm tracker's own state, the fitted pose removes 71% of the squared speed error that regressor leaves, the pre-fit pose about a fifth and a stranger's 1%. At 10 cm the shares are 21% and under 2%.

We therefore reran both pose arms at 30 cm on the pre-fit poses, which see only the points inside the box. Files, agents, tracks and the stranger assigned to each pedestrian are unchanged, and the two shuffled arms agree (+0.40%, p=0.54). Own pose now beats a stranger's by 1.34% (p=0.10, CI $[-3.1, +0.4]$) and does not beat no pose (+0.36%, p=0.54). At fixed epochs 26, 28 and 30 it reads −1.4, −2.6 and −1.5%, significant at the second only (p=0.02). The fit accounts for 3.98 of the 5.32 points (p=0.0001, between 3.5 and 4.9 on every seed), and on walkers −6.8% becomes −0.8% (p=0.36).

What survives appears to follow the sensor. Split by mean LiDAR returns over the observed second, the pre-fit content effect is −0.3% below 30 returns, −1.0% and −1.4% in the two middle strata, and −3.47% (p=0.010, CI $[-5.6, -1.4]$) on the 2,365 pedestrians with at least 200, where it also beats no pose (−1.99%, p=0.041). The top stratum is better than the bottom on every seed, by 3.1 points (p=0.004), and at every fixed late epoch it reads −3.2 to −4.6%. The bottom stratum is less settled. At those epochs it reads −1.4 to −2.1% (n.s.), so the gap between the two narrows to 1.7 to 2.5 points. The stranger arm is also 0.9% worse on pre-fit than on fitted poses in the top stratum (p=0.015), and against the fitted stranger the effect is −2.6% (p=0.03). The fitted poses show no gradient (−5.3, −5.4, −4.1 and −6.0%, p=0.41), as a leak that does not depend on the sensor would not. On the offboard track the same stratum is null (Sec. IV-D). In the example of Fig. 3 below 30 returns the points lie between thigh and shoulder, with none on the head or the lower legs. In the one with at least 200 they cover the body from head to foot.

It follows the tracker's error, not the manoeuvre. Table III splits the 2,272 well-resolved pedestrians with at least 1 s of future two ways. By what the pedestrian goes on to do, the four manoeuvre measures of Sec. IV-D, own pose beats the stranger's in all 16 quartiles (−1.2 to −6.0%) and no measure shows a trend: top and bottom quartile differ by −0.4 to +2.6 points ($p \geq 0.19$). By how wrong the tracker is at prediction time, the effect is −1.5% in the quartile with the smallest velocity error and −4.6% in the largest, a difference of 3.1 points (p=0.001), and −0.3% against −5.8% over position error (5.5 points, p=0.001). The detection noise is drawn independently of the pedestrian, and the trend holds on complete one-second histories alone (2.3 points, p=0.004), so it is not the filter's cold start. On all pedestrians it is 2.3 points over position error (p=0.008) and 1.4 over velocity error (p=0.09). Against no pose, own pose is −2.1% on the 2,272 (p=0.039) and −4.0% on the half with the larger velocity error (p=0.003). The fitted poses do the opposite. Their gain does not follow the tracker's error (p=0.08 and 0.87) and shrinks with the manoeuvre, from −8.4% to −3.5% over constant-velocity residual (p=0.013), as a leaked speed is worth most where the pedestrian keeps it. Fig. 4 shows four pedestrians from the quartile with the largest tracker error.

Three caveats bound this. The tracker is simulated, with Gaussian detection noise rather than a detector's correlated, occlusion-driven errors. The pre-fit poses are per-frame and unsmoothed, so a prior fitted to the body alone might recover more, though on the 347 frames with labelled keypoints [20], [21] the fitted poses are no more accurate than the pre-fit ones (92.7 against 87.6 mm MPJPE). And returns are one of two stratifications we scored at this noise level, four strata each, so p=0.010 is uncorrected over eight strata, and the splits

TABLE III

WHERE THE PRE-FIT GAIN SITS. CONTENT EFFECT IN PERCENT AT s=30 CM ON THE 2,272 PEDESTRIANS WITH AT LEAST 200 RETURNS AND 1 S OF FUTURE, BY QUARTILE OF EACH MEASURE (Q1 LOWEST, 568 PEDESTRIANS EACH), n=5 SEEDS. TREND IS Q4 MINUS Q1 IN POINTS WITH ITS PAIRED p. ASTERISKS MARK p<0.05, UNCORRECTED.

	Q1	Q2	Q3	Q4	Trend (p)
Tracker error at prediction time					
Velocity	−1.5	−2.0	−5.8*	−4.6*	−3.1 (0.001)
Position	−0.3	−5.1*	−2.6*	−5.8*	−5.5 (0.001)
Manoeuvre over the labelled 8 s future					
Constant-velocity residual	−4.6*	−2.5	−1.3	−4.9*	−0.4 (0.66)
Heading variation	−4.7	−2.2	−4.7*	−2.3	+2.4 (0.37)
Speed range	−6.0*	−4.5*	−1.2	−3.4*	+2.6 (0.19)
Peak acceleration	−4.8*	−2.6	−2.4*	−4.5*	+0.3 (0.86)

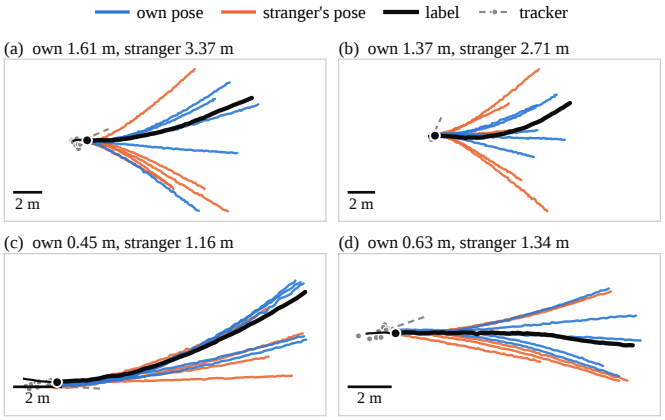


Fig. 4. Four pedestrians on which own pose helps under the 30 cm tracker, drawn in the label frame at prediction time. Each coloured line is the best of six modes of one seed at the final epoch, and titles give the mean minADE. The examples are selected on the effect. Among the 568 well-resolved pedestrians with the largest tracker velocity error, own pose is better on all five seeds for 43 and worse on all five for 28 at that epoch, and these are the four largest gains with a complete future. In the quartile with the smallest error the counts are 33 and 32. Dots are the tracked history, and the dashed line extends the tracker's velocity over 3 s.

of Table III were chosen after that result and subdivide its stratum. What the two runs establish is that the same body adds nothing to an offboard track and that a pose fitted to a labelled track inherits the label. That it adds to a causal track where the sensor resolves it rests on one stratification.

V. WHY POSE DOES NOT PAY ON OFFBOARD TRACKS

A. What the oracle ladder measures

The content test bounds what pose is worth, and the oracle of Sec. IV-C shows the branch can carry information. Neither says what kind of information this pathway turns into accuracy, so we degraded the oracle.

Each level of Table IV feeds the branch the same direction-only oracle, the bearing of the agent's own future displacement at 3 and 8 s, corrupted by a fixed angular error. Each is labelled by the median bearing error actually delivered: 0.6745σ by construction, since the corruption is a fixed per-agent rotation, and measured to within 0.7° of it on 11,328 scored agents. Displacement magnitude is withheld, so the channel carries direction alone. The error is drawn once per pedestrian, not per epoch, as a real estimator's error is a property of the observation.

At 25k scenes every level converts, and the gain stays large: -34.50% with an exact bearing, -19.77% at 5.0°, -15.68% at 10.0° and -13.57% at 15.1°, all at $p \leq 0.0016$ against minimum detectable effects of 1.4–2.9 pp. The decay is not a single exponential. Two terms fit it to 0.4 pp:

$$\Delta minADE(\theta) \approx -\left[\,13.1\% + 21.4\% \cdot e^{-\theta/4.4^\circ}\,\right], \quad (1)$$

a precision term with a half-life of 3.0° over a term taken as flat across the fitted span, an assumption the half-life depends on. That term is slow, not permanent, and two levels held out of the fit measure it. By 24.9° the precision term is below 0.1%, so that level reads the slow term almost alone, and it still converts: -10.39% (p=0.003, CI $[-12.9, -7.9]$) against the -13.2% Eq. (1) extrapolates (Table IV). At 44.4° the gain is -6.92% (p=0.013, CI $[-10.3, -3.5]$), so about half of it is left.

The angular axis is not an acceptance threshold. It is tempting to read θ as the accuracy a cue must reach. The ladder refutes that reading itself. The baseline already receives the labelled box heading, as a per-timestep sine–cosine pair for every agent (Fig. 2), and that heading estimates the same future displacement bearing at 3.9° in median, better than the 15.1° channel at every percentile (9.3° against 25.8° at the third quartile, 20.1° against 36.3° at the ninetieth). A channel uniformly less accurate than an input the model already has still buys -13.6%. Nor does the branch pay for merely being occupied: carrying a stranger's pose it costs $\sim$1.6% (Sec. IV-B). The 34.5% is content, and the only content this channel has is the realised future.

The surviving term is future-side mode selection. Eq. (1) splits the gain, and the two terms behave differently in ways that identify them. Per-agent records at each run's own best epoch, paired between arms on 16,486 agents and averaged over seeds, put the persistent term at the end of the trajectory: refitting the decomposition at each horizon leaves the precision half-life at 2.9–3.9° throughout, while the persistent share of the gain rises from 21% of the 2 s error reduction to 39% at 8 s and 53% at the endpoint. The selection reading predicts more: corrupting the bearing should barely affect which of the six modes is right. It does not. The gap between the top-ranked mode and the best of six, which is what the forecaster loses by choosing wrongly, falls from 0.67 m to 0.26 m with an exact oracle, and still to 0.29 m at 15.1°: the corrupted channel closes 92% of the perfect oracle's selection gap while recovering 39% of its minADE. A coarse future bearing says which way the agent went, and only a precise one refines where along that way, which is what 3° of error destroys.

minADE takes the best of six modes whatever their rank, so selection reaches it only through the decoder's scoring step, which keeps six of 64 intention queries: a better ranking changes which six survive. That is also where a pose effect could hide from our headline metric, so we scored the ranking

columns directly on the 25k arms. They are null as well: against a stranger's pose, top-1 ADE is -2.07% (p=0.11, MDE 3.7 pp), best-mode rank -0.93% (p=0.22) and endpoint mode spread $+0.17\%$ (p=0.53), over the same 16,486 agents.

A null at low power looks like a threshold. At 579 scenes this ladder read as a detection floor, significant at $10.0°$ and not at $15.1°$ (-1.4%, p=0.47), where the 25k ladder gives -13.6%. That level's MDE at 579 scenes was 6.7 pp: the floor was the angle at which three to five seeds went blind.

B. Why body facing does not pay

The oracle converts a bearing read off the realised future. Body facing is not that: it is an observation-time estimate of a quantity the model already measures. Three properties of it, on the population the ladder scores, say what is left for it to contribute.

It is a noisier copy of an existing input. The box heading estimates the future displacement bearing at $3.9°$ (+3 s) and $7.1°$ (+8 s) in median, while SMPL facing reaches $12.3°$ (n=5,824) and $15.1°$ (n=6,206). Box heading is the better estimate at every percentile we measured, the two differ by $10.0°$ in per-agent median, and both derive from the same returns. We calibrated the SMPL forward axis against the box heading before measuring either.

Most of its accuracy is the trajectory restated. 95.5% of the scored agents move, and for a mover the pipeline snaps root orientation to the direction of travel whenever the raw estimate deviates by more than $90°$ (Sec. III). Movers sit at $11.9°$, while the 262 stationary agents, the only ones whose facing their velocity does not already imply, sit at $41.2°$. The pooled median largely measures the snapping.

Where it would be non-redundant, it is worst. On the one case where velocity carries no directional information (which way a stationary pedestrian steps off), SMPL facing predicts onset direction at roughly twice chance over the 1,705 onset events among 19,939 stationary validation pedestrians: median error $44.2°$, 50.6% within $45°$ (chance $90°$, 25%). That is informative, and not an artifact of our axis fitting: the box heading estimates the same quantity at $25.2°$, and paired on the same events SMPL is $7.3°$ worse in median (Wilcoxon p=2.6×10^{-43}). Nor does pose take over where perception degrades: over those events the gap favours the box heading in all ten testable strata ($p < 10^{-5}$ each) and widens with range, to $+32.9°$ at 41–75 m, because both derive from the same returns.

Facing is therefore redundant where it is accurate and inaccurate where it would be non-redundant, and neither half of that squeeze is the selection signal the oracle trades on. What survives it is what the content test measures directly: $+0.04\%$ at 25k scenes, bounded to $\pm1.7\%$.

Removing the redundant input does not rescue it. Redundancy says why little is left for facing to add, but it is not by itself a mechanism that blocks conversion. So we deleted the box-heading channel, retrained all six cells and repeated the content test, at 579 scenes and eight seeds. Nothing appears. The difference in content effect between the two conditions is +2.9 pp on the 13,006 moving agents, the wrong sign for a benefit that redundancy had been masking, and -0.6 pp on the stationary ones (p=0.65). Deleting the channel costs the no-pose baseline only 1.5% on movers, because velocity stays in the input, so there was never much room for facing to fill.

TABLE IV

THE ORACLE LADDER AT 25K SCENES. A DIRECTION-ONLY ORACLE CORRUPTED BY A FIXED PER-AGENT ANGULAR ERROR, EACH LEVEL LABELLED BY THE MEDIAN BEARING ERROR ACTUALLY DELIVERED. TESTS ARE PAIRED t AGAINST EACH SEED'S OWN NO-POSE RUN, AND NEGATIVE FAVOURS THE ORACLE. MDE IS THE MINIMUM EFFECT A LEVEL COULD DETECT AT 80% POWER. EVERY LEVEL CONVERTS, OUT TO $44.4°$. THE LAST TWO LEVELS ARE HELD OUT OF EQ. (1).

Bearing error	n	ΔminADE	p	95% CI	MDE
$0.0°$	3	-34.50%	0.0001	$[-35.6, -33.4]$	1.4
$5.0°$	3	-19.77%	0.0007	$[-22.1, -17.5]$	2.9
$10.0°$	3	-15.68%	0.0009	$[-17.7, -13.6]$	2.6
$15.1°$	3	-13.57%	0.0016	$[-15.9, -11.2]$	2.9
$24.9°$	3	-10.39%	0.0031	$[-12.9, -7.9]$	3.1
$44.4°$	3	-6.92%	0.0127	$[-10.3, -3.5]$	4.2

A frame-rotation control, not a degraded heading. The heading the baseline receives is an offboard label too (Sec. IV-E), so it may be better than a deployed tracker's, and facing might pay only where it is worse. Deleting the channel cannot test that: MTR's agent-centric frame, and the intention anchors defined in it, stay aligned to the clean heading. We rotated that frame by a fixed per-agent error $\mathcal{N}(0, \sigma^2)$, which frame, anchors and map inherit, and repeated the content test at 25k scenes and three seeds (Table II). The no-pose baseline loses 5.8% at σ=$15°$ and 12.6% at $30°$, and pose recovers none of it: -0.14% (p=0.90) and -0.11% (p=0.93) against shuffled. This does not test a degraded heading, for two reasons. The per-timestep heading channel stays the clean heading expressed in the rotated frame, so the model can read the error we added. And the pose root stays in world coordinates, so the body could not have stated the correction. The control measures the cost of a misaligned frame, and whether facing pays under an onboard-quality heading is untested here. The clean condition reproduces our main cells to within 0.4%.

C. A general condition

Two results locate where pose could have helped. At 2 s, minADE is 0.097 m at 25k scenes, and a perfect future-direction oracle through the same branch removes only 1.4 cm of it (-14.7%): in absolute terms there is almost nothing for any cue to win. At long horizons the residual is dominated by destination and intent, which body configuration does not disclose.

In general form, and offered as a hypothesis from one modality on one dataset rather than a law: *an auxiliary modality improves a forecast only where the baseline's own residual leaves room in absolute terms, where that residual is not dominated by intent the modality cannot observe, and where the modality carries information about the cue that the model's existing inputs do not already carry.* The third

clause is not a comparison of marginal accuracies: Sec. V-A converts a channel less accurate at every percentile than an input the model already has. What disqualifies facing is that its error is shared with the input it would supplement (Sec. V-B). Sec. IV-E tests the clause from the other side, and shows it has a trap. A cue whose pipeline was fitted to the input it supplements carries that input, and looks informative exactly when the input is degraded. Redundancy is a property of a modality together with the inputs beside it, and an offboard-labelled benchmark sets those inputs at offboard quality.

VI. Practice and Limitations

For an AV stack, three consequences follow. First, before integrating an auxiliary modality, measure how much of its cue it explains beyond the inputs the model already receives, take those inputs at the quality a vehicle will have, and ask what the cue's own pipeline was fitted to. None of this costs training. A simulated causal tracker is a few lines in a data loader, and our fitted poses predicted the labelled speed with $R^2{=}0.94$, and with 0.78 on moving pedestrians seen through fewer than 30 returns, which would have flagged the leak before any run. Second, an oracle ladder is worth its handful of runs as a positive control and an upper bound on the pathway, if its MDE is below the effect you would act on (Sec. V-A). Third, the cost of an extra branch ($\sim$1.6% here, at the best-epoch operating point) is easily mistaken for a modality effect in either direction. A shuffled-pose control separates the two, and early fusion avoids the cost while still reading the channel.

Limitations. We test one forecaster, one dataset and one pose estimator, the model sees pedestrians and map only, and we do not model pose–pose interaction. The largest scale we train is 25k of the 299,220 exported scenes, a compute limit, and the ladders run at $n{=}3$ per level, five for track noise. We did not test a degraded heading: the frame-rotation control leaves the clean heading readable, and the default pose arm carries its root in world coordinates (Sec. V-B). The onboard track is simulated (Sec. IV-E). The pre-fit poses are still estimated from points cropped by offboard boxes, the pre-fit control covers the 30 cm level only, and we have not run either through early fusion. The redundancy counterfactual is run at 579 scenes rather than at 25k. WOMD ships camera data only as quantised tokens, so a camera-based pose or gaze estimate cannot be tested at this scale.

VII. Conclusion

Body pose, recovered from LiDAR, does not improve pedestrian trajectory forecasting on WOMD's offboard tracks. The null is controlled against a shuffled-pose baseline and holds across scale, representation and integration, and an oracle through the same branch rules out a dead pathway. Under a simulated onboard tracker own pose beats a stranger's by 5.3%, and three quarters of that is the offboard label, carried in by a pose pipeline that fits the gait to the track. Poses that never saw the track gain a non-significant 1.3% overall and 3.5% on the best-resolved bodies, the latter in one stratification, where it grows with the tracker's error and with no manoeuvre. The results are one mechanism: a cue pays only for what it carries beyond the model's existing inputs, an offboard track leaves the body nothing to add, and a cue fitted to a label carries the label.

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